

SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE
SUBJECT - Data Analysis SAS / SPSS / R

Aim - Applying basic data cleaning functions: handling missing values using `na.omit()/replace_na()` in R. import dataset.

Input :

The image displays two screenshots of the RStudio environment, showing the progression of an R script for data cleaning.

Top Screenshot: The script includes the following code:

```
1 install.packages("stringr")
2 install.packages("tidyr") # for separating columns after splitting
3 library(stringr)
4 library(tidyr)
5 library(dplyr)
6
7 # =====
8 # 1. IMPORT DATASET
9 # =====
10 states_df <- data.frame(State = state.name)
11
12 print("--- Original Dataset ---")
13 head(states_df)
14
15 # =====
16 # 2. USING str_sub() (Substring)
17 # =====
18 # Scenario: Extract specific parts of the state name
19 # Example A: First 3 letters as a code
20 states_df$State_Code <- str_sub(states_df$State, 1, 3)
21
22 # Example B: Last 2 letters of the state name
23 states_df$State_End <- str_sub(states_df$State, -2, -1)
24
25 print("--- Data after str_sub() ---")
26 head(states_df)
27
28 # =====
29 # 3. USING str_split() (Split String)
30 # =====
31 # Scenario: Some states have two words, e.g., "New York"
32 # We want to split each state name into separate words
33
34 # Method A: Basic split (returns a list)
35 split_list <- str_split(states_df$State, " ")
36 print("--- First state's split (List format) ---")
37 print(split_list[1])
38
39 # Method B: Split and assign to new columns (works for states with up to 2 words)
40 split_matrix <- str_split(states_df$State, " ", simplify = TRUE)
41 states_df$word1 <- split_matrix[, 1]
561
```

The Environment pane on the right shows the following objects:

Object	Class	Size	Modified
starts_with_	32 obs. of 2 variables		
state.center	List of 2		
state.x77	num [1:50, 1:8]	3615 365 2212 ...	
states_df	50 obs. of 5 variables		
strong_raid_	12 obs. of 19 variables		
survivors_o_	422 obs. of 12 variables		
tidy_states	50 obs. of 6 variables		
titanic	891 obs. of 12 variables		
young_high_	62 obs. of 12 variables		
young_nasse	71 obs. of 12 variables		

Bottom Screenshot: The script continues with the following code:

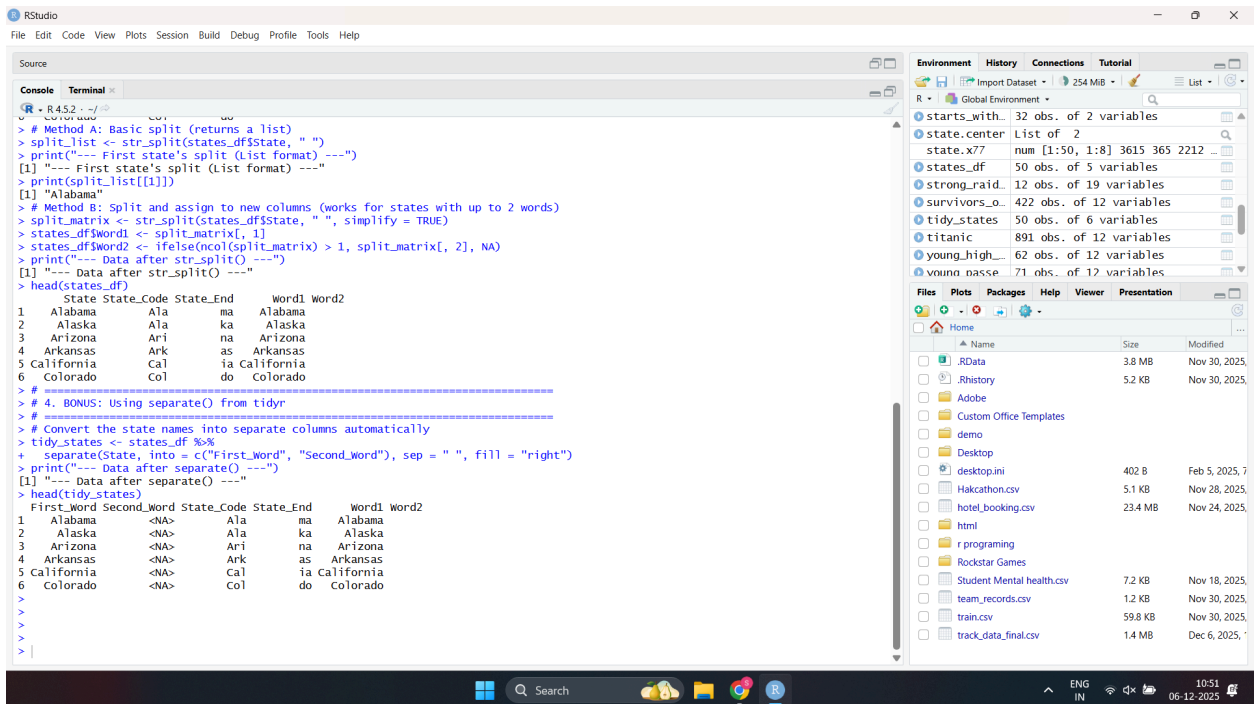
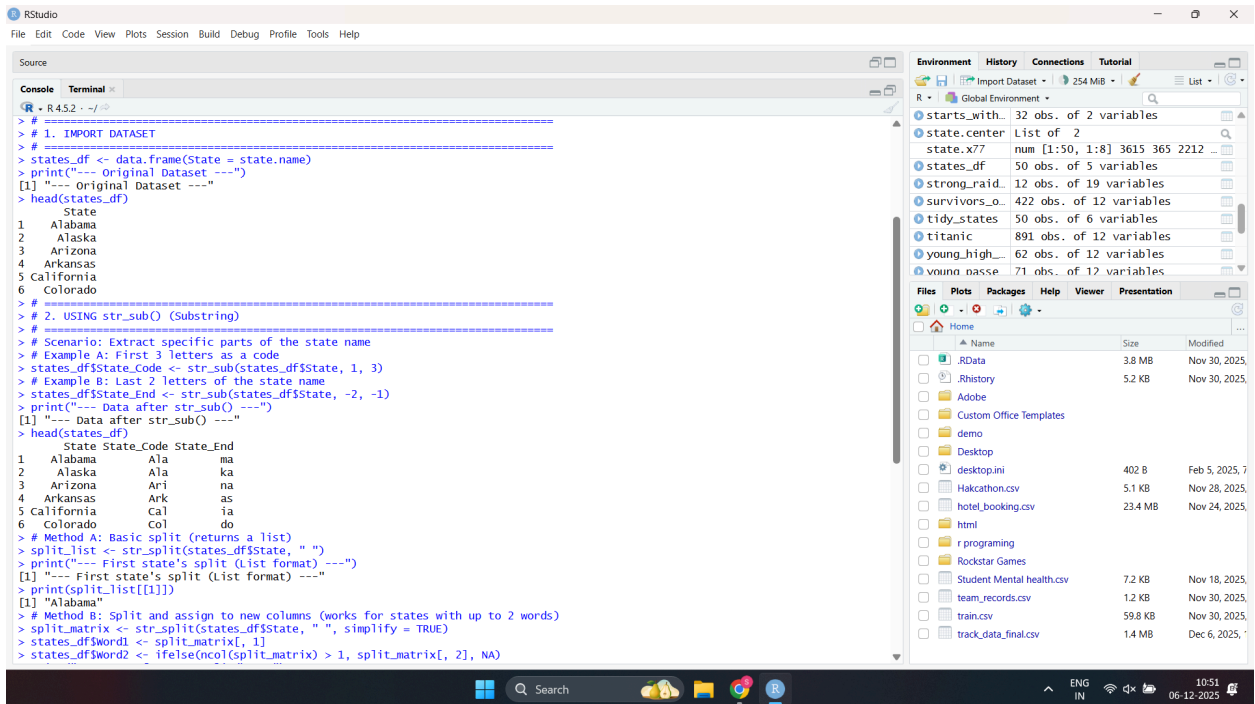
```
17 # =====
18 # Scenario: Extract specific parts of the state name
19 # Example A: First 3 letters as a code
20 states_df$State_Code <- str_sub(states_df$State, 1, 3)
21
22 # Example B: Last 2 letters of the state name
23 states_df$State_End <- str_sub(states_df$State, -2, -1)
24
25 print("--- Data after str_sub() ---")
26 head(states_df)
27
28 # =====
29 # 3. USING str_split() (Split String)
30 # =====
31 # Scenario: Some states have two words, e.g., "New York"
32 # We want to split each state name into separate words
33
34 # Method A: Basic split (returns a list)
35 split_list <- str_split(states_df$State, " ")
36 print("--- First state's split (List format) ---")
37 print(split_list[1])
38
39 # Method B: Split and assign to new columns (works for states with up to 2 words)
40 split_matrix <- str_split(states_df$State, " ", simplify = TRUE)
41 states_df$word1 <- split_matrix[, 1]
42 states_df$word2 <- ifelse(ncol(split_matrix) > 1, split_matrix[, 2], NA)
43
44 print("--- Data after str_split() ---")
45 head(states_df)
46
47 # =====
48 # 4. BONUS: Using separate() from tidyr
49 # =====
50 # Convert the state names into separate columns automatically
51 tidy_states <- states_df %>%
52   separate(State, into = c("First_Word", "Second_Word"), sep = " ", fill = "right")
53
54 print("--- Data after separate() ---")
55 head(tidy_states)
56
57
```

The Environment pane on the right shows the same objects as the top screenshot, plus the new `tidy_states` object.

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Output :



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